CSA0619-Design and Analysis of Algorithm

28. Brute Force String Matching Algorithm:

Analysis:
The Brute Force String Matching algorithm compares the pattern with the text character by character. If a mismatch occurs, the pattern is shifted by one position and compared again. It is simple but performs many repeated comparisons, making it slow for large log files.

Optimization:
The Brute Force algorithm performs many repeated comparisons when searching large log files. As the size of the text increases, the search becomes slower. Algorithms such as **KMP**, **Boyer–Moore**, and **Rabin–Karp** reduce unnecessary comparisons and provide faster searching.

Report Analysis

Case	**Time Complexity**
Best	O(m)
Average	O(n × m)
Worst	O(n²)

Space Complexity: O(1)

Where:

- **n = Length of the text**
- **m = Length of the pattern**